

Beyond Epistemic Collapse: Disagreement-Aware Scientific Retrieval-Augmented Generation

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Abstract

Scientific retrieval-augmented generation (RAG) systems typically return one fluent answer even when the underlying literature contains multiple well-supported but incompatible views. We call this failure mode *epistemic collapse*: compression of structured disagreement into a single response. We present EVIRAG (Evidence-centric, Viewpoint-Inclusive RAG), a framework for scientific question answering that combines adversarial retrieval, claim-level disagreement graphs, causal disagreement attribution, multi-view synthesis, and evidence-structured confidence labels. We also introduce EVIRAG-BENCH, a 1,250-query benchmark spanning five contested scientific domains. Across eight systems and three controls, EVIRAG substantially outperforms vanilla RAG on ContradictionRecall and dense-encoder ViewpointCoverage while reducing Single-View Concentration. A single-call structured-prompt baseline recovers most of the viewpoint-coverage gap but only about a quarter of the contradiction-recall gap, isolating what the claim graph contributes beyond output formatting. Human evaluation with three annotators confirms markedly higher epistemic completeness for EVIRAG (Krippendorff’s $\alpha=0.72$, $p<0.001$, Cohen’s $d=2.71$), an ordering that survives censoring the rubric at its second-highest anchor. An offline precomputed graph reduces warm-query latency by over an order of magnitude. These results support treating disagreement preservation as a primary design objective for disagreement-heavy scientific QA.

1 Introduction

Scientific question answering has advanced rapidly through retrieval-augmented generation (RAG), which grounds model responses in

retrieved documents (Lewis et al., 2020). Yet a fundamental challenge remains: when the literature is genuinely contested, what should a responsible system return? Current architectures optimize for a single coherent answer—a design suited to lookup queries but structurally misaligned with contested knowledge, where authoritative sources support mutually incompatible conclusions. Questions such as *Does homework improve achievement?* (Cooper et al., 2006; Masalimova et al., 2023; Scheb, 2023), *Do statins help in primary prevention?* (Ridker et al., 2008), or *Does raising the minimum wage reduce employment?* (Card and Krueger, 1994) rest on large but internally contested literatures, yet standard RAG returns one answer optimized for fluency and local support—an answer that can be well-written and source-grounded while still erasing materially important disagreement.

Epistemic collapse. We call this failure mode *epistemic collapse*:¹ conversion of structured multi-view evidence into one response. Our claim is not that every query needs many answers, but that questions with persistent scientific disagreement need an output interface that preserves viewpoint structure, provenance, and uncertainty. Improving retrieval recall alone does not solve the problem if synthesis still compresses the evidence into one paragraph.

Approach. EVIRAG uses adversarial retrieval to surface competing evidence, constructs a claim-level disagreement graph, assigns causal disagreement types with CDA-7, tracks temporal drift, and generates a multi-view answer with evidence-

¹We use “epistemic collapse” as a technical term for the failure mode described here; the term has different meanings in modal epistemology and should not be confused with that usage.

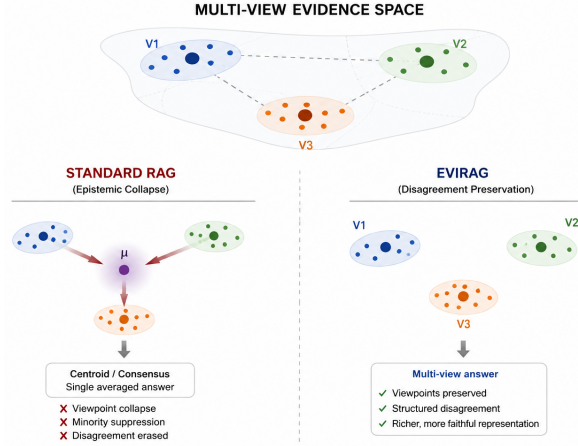


Figure 1: Conceptual contrast between epistemic collapse in Standard RAG and disagreement preservation in EVIRAG. **Left:** Standard RAG compresses multiple distinct viewpoints (V1, V2, V3) into a single consensus centroid (μ), resulting in viewpoint collapse and erased disagreement. **Right:** EVIRAG preserves all viewpoints in a structured multi-view answer, providing a richer and more faithful representation of the evidence space.

structured confidence (Figure 1). Instead of asking the model to resolve controversy prematurely, it asks the model to represent controversy faithfully.

Contributions.

- We give a typed formalization of epistemic collapse for scientific RAG and introduce four viewpoint-preservation metrics: Single-View Concentration (SVC), Epistemic Divergence (ED), Polarization Index (PI), and Epistemic-Coverage@K (EC@K).
- We present EVIRAG, a seven-stage architecture combining adversarial retrieval, claim-graph reasoning, CDA-7 disagreement attribution, temporal drift tracking, multi-view synthesis, and evidence-structured confidence.
- We introduce CDA-7, a seven-class causal disagreement taxonomy, and a four-way controversy typology that calibrates confidence to the structure of the retrieved evidence.
- We introduce EVIRAG-BENCH, a 1,250-query benchmark spanning five contested scientific domains, evaluated against ablations and external baselines with automatic metrics and human epistemic completeness ratings.
- We report a set of controls that separate our architecture from cheaper explanations of its gains—a closed-book baseline, a budget-matched retrieval control, and a single-call

structured-prompt baseline—together with a per-system accounting of LLM calls and tokens (§5.6), together with a direction-validity audit of our own metric suite that reports a negative result: reference-free measures of viewpoint concentration are not direction-valid (§3.3).

2 Related Work

RAG, verification, and conflict handling. Standard RAG optimizes retrieval quality, attribution, or factual precision for a single answer (Lewis et al., 2020; Gao et al., 2023; Min et al., 2023; Ru et al., 2024), and scientific QA systems such as SciFact, SciFact-Open, and PaperQA2 justify claims against evidence while still collapsing the response to one answer (Wadden et al., 2020, 2022; Lála et al., 2023; Skarlinski et al., 2024). Work on conflicting evidence and multi-answer QA recognizes that sources diverge but aims to resolve the conflict or pick the best surviving answer (Zhong et al., 2023; Wang et al., 2025; Xie et al., 2024); ConflictQA (Xie et al., 2024), for instance, offers neither causal attribution nor per-view confidence. EVIRAG *instead makes preserving disagreement the primary objective*.

MADAM-RAG vs. EVIRAG. The closest work is MADAM-RAG (Wang et al., 2025), which runs a structured debate (Proposer → Opposer → Moderator) to *resolve* conflicting evidence into one answer. EVIRAG shares the multi-agent architecture but inverts the objective: it *preserves* conflict. The distinction shows in the results—MADAM-RAG reaches CR= 0.618 against EVIRAG Full’s 0.742, a gap holding across all five domains—because its Moderator collapses viewpoints at synthesis exactly as a single-agent baseline does, merely after a richer debate. Neither design dominates: a resolved answer suits tasks needing one verdict, while EVIRAG’s *format suits preserving disagreement for human deliberation*.

Multi-perspective generation and epistemic uncertainty. Multi-perspective summarization produces diverse summaries from opinionated corpora (Bražinskas et al., 2020; Hayashi et al., 2021) but supplies no scientific provenance, disagreement typing, or evidence-grounded confidence. Work on LLM uncertainty studies how confidently models express statements (Xiong et al., 2024; Kadavath et al., 2022); we target the epistemic

structure of the retrieved corpus instead. CDA-7 addresses a gap in NLI-based claim verification: standard NLI, and claim-level entailment approaches such as FActScore (Min et al., 2023), label a pair *contradicts* or *supports* without explaining *why*. CDA-7 adds that causal dimension, motivated by philosophy-of-science accounts of how disagreements are sustained (Longino, 2002; Solomon, 2001). Table 12 (Appendix D) summarises the positioning.

3 Epistemic Collapse: Formalization

3.1 Typed objects

Table 1 fixes the internal structure of every object the paper manipulates; leaving these abstract, as earlier presentations did, makes the metric definitions ambiguous. An *embedding* throughout is a sentence embedding from all-MiniLM-L6-v2 (Reimers and Gurevych, 2019), written $\phi(\cdot)$, so the space is $\mathbb{S}^{383} = \{x \in \mathbb{R}^{384} : \|x\|_2=1\}$: vectors are L_2 -normalized *before* any centroid is formed, and every centroid is a renormalized mean of unit vectors. All cosines are therefore inner products of unit vectors on a common scale.

3.2 Collapse

Definition 1 (Epistemic Collapse). *A system S exhibits epistemic collapse on query q with ground-truth view set V^* if S returns a_{sv} such that $VC(a_{sv}, V^*) \leq 1/|V^*|$, where VC is Viewpoint-Coverage under a fixed similarity threshold.*

Collapse is behavioral, not an interface property. Definition 1 concerns measured coverage, not whether the system emitted one paragraph or several. We therefore do not claim epistemic collapse is categorically distinct from incomplete summarization: it is what incomplete summarization looks like when the omitted material is *structured disagreement*—a special case with a specific cause, cost, and remedy. A long single answer that does lay out several views does *not* collapse here; our structured-prompt baseline (§5.6) is exactly such a system. The empirical claim is correspondingly narrow: standard RAG collapses by *default* (vanilla RAG: $VC_{emb}=0.423$), and the residual gap after prompting is where explicit structure earns its cost.

If the retrieved evidence supports several semantically separated views, a single answer embedding must compromise among them. Topical re-

Object	Type
Document d_i	(τ_i, m_i) : token sequence and metadata tuple (year, venue, sections); segmented into 256-token passages, 32-token overlap
Claim c_j	(t_j, π_j, e_j) : declarative sentence asserting exactly one checkable proposition; provenance pointer (doc, passage, span); $e_j = \phi(t_j)$
View v	(σ, C_v, ω) : summary; supporting claims $C_v \subseteq C$; weaknesses and inherited provenance. Centroid c_v is the renormalized mean of $\{e_j : c_j \in C_v\}$
Graph G	(C, E) with $E \subseteq C \times C \times L$ and label set $L = \{\text{SUPPORTS, CONTRADICTS, NEUTRAL}\}$; each CONTRADICTS edge additionally carries a CDA-7 cause label
Answer a_{sv}	token sequence with induced claim set $C(a_{sv})$
<i>Structure each metric consumes</i>	
VC, SVC, ED	embeddings e_j , centroids only
CR	labeled claim pairs in E
PI	signed adjacency of G
Synthesis, FS	provenance pointers π_j

Table 1: Typed objects and the structure each metric requires (CCS, like VC and SVC, uses embeddings only). D , C , and V denote the corpus, claim set, and view set; V^* is the gold view set. Where a statement genuinely treats elements abstractly—notably the cardinalities in Definition 1—no internal structure is assumed.

trieval can amplify this effect by oversupplying the dominant view, creating what we call a *consensus attractor*. Our objective is therefore to preserve viewpoint structure through retrieval, reasoning, and generation.

Remark 1. *Let $c_1, \dots, c_k$ be unit-normalized ground-truth view centroids. If $\|c_i - c_j\|_2 > 2\delta$ for all $i \neq j$, then any single answer embedding a can satisfy $\|a - c_i\|_2 \leq \delta$ for at most one i .*

Justification. If a were within δ of both c_i and c_j , the triangle inequality would give $\|c_i - c_j\|_2 \leq \|c_i - a\|_2 + \|a - c_j\|_2 \leq 2\delta$, contradicting the assumption. The remark’s scope is narrow: it concerns a *single point* in embedding space, and a real long-form answer is not one—an answer enumerating several views induces several claim embeddings and escapes the bound. The remark does not show a single-answer *interface* must collapse; it shows why single-vector summarization objectives, which RAG synthesis prompts implicitly approximate, press toward the dominant view when gold views are well separated. The paper’s empirical claims rest on §5.

3.3 Metrics

Single-View Concentration (SVC). SVC measures how unevenly a response distributes claim mass across a query’s *gold* views. Each claim $c_j \in C(a)$ is assigned to the nearest gold view by cosine, $\arg \max_i \langle \mathbf{e}_j, \mathbf{c}_{v_i^*} \rangle$; let p_i be the resulting fraction of claim mass on v_i^* . SVC is one minus the normalized Shannon entropy of p :

$$\text{SVC}(a, V^*) = 1 + \frac{\sum_{i=1}^{|V^*|} p_i \log p_i}{\log |V^*|} \quad (1)$$

with $0 \log 0 = 0$. $\text{SVC}=1$ means all claim mass sits on one gold view (the collapse endpoint); $\text{SVC}=0$ means uniform spread. **Lower is better.** Being reference-based and assignment-based, SVC depends on the *argmax* of a cosine rather than its magnitude, so it is invariant to the absolute-cosine inflation that encoder anisotropy induces. We report SVC throughout §5.

Why SVC is reference-based. A concentration metric would be more useful if it needed no gold annotation, but cluster geometry alone cannot supply one. The natural reference-free construction—the Consensus Collapse Score (CCS), comparing a response’s dominant claim-cluster centroid against the mean of its own cluster centroids—is **direction-invalid, which we report as a negative result**. A response spread over several well-separated views necessarily pushes $\mathbf{c}_{\text{dom}}$ away from that mean, so CCS *rises* with viewpoint diversity even as concentration falls. The failure is systematic, not anecdotal: CCS orders 0 of 8 gold instances correctly over 6 domains and 2 annotation sources ($p=0.008$), while every reference-based metric orders all 8 correctly (Appendix F). The lesson generalizes—a response’s own cluster geometry cannot separate balanced coverage from tight clustering without an external reference—and it is why SVC scores against V^* . CCS appears in the appendix for auditability only.

Other metrics. **Epistemic Divergence (ED)** is the mean pairwise cosine distance among view centroids, so higher values mean a more fragmented evidence landscape. **Polarization Index (PI)** is a signed-Laplacian measure of conflict separation in G (Traag and Bruggeman, 2009), capturing whether G decomposes into opposed clusters (high) or a gradual spectrum (low). **EpistemicCoverage@K (EC@K)** is the fraction of gold viewpoints present in the top- K retrieved

chunks. Because it scores the retrieved pool rather than the response, we define it as a retrieval-side diagnostic—it localizes whether a coverage failure originated before synthesis—and not as a system-ranking metric; it is part of the released metric suite but is not among the metrics compared in §5. ED and PI likewise serve an internal role, defining the controversy typology below.

Evaluation metrics. **ContradictionRecall (CR)** is the fraction of gold-annotated contradictory claim pairs that a system recovers in its output; it adapts the standard precision/recall paradigm from scientific claim verification (Wadden et al., 2020). **ViewpointCoverage (VC)** is the fraction of annotated ground-truth viewpoints surfaced in the response, measured by dense-encoder cosine similarity (VC_{emb} ; all-MiniLM-L6-v2 (Reimers and Gurevych, 2019), threshold $\theta=0.65$), following the coverage framing of Malaviya et al. (2024). A keyword-overlap variant (VC_{kw}) is retained only as a sanity check in Appendix F: it saturates between 0.892 and 0.991 across all compared systems and consequently does not discriminate between them, so we do not report it as a headline metric. **CCE** (Confidence Calibration Error) is the mean absolute deviation between the system’s emitted confidence tier and the gold controversy-class label; lower CCE indicates better-calibrated uncertainty signals. CCE is only defined for systems that actually emit a confidence signal, so we report it for those systems alone rather than crediting silence as calibration (§5.5). **FaithfulnessScore (FS)** follows FActScore-style NLI verification (Min et al., 2023): each atomic claim in the response is verified against its cited source chunks and the fraction supported is reported. CR, VC_{emb} , SVC, CCE, and FS are the five metrics used throughout §5.

Controversy typology. The (ED, PI) plane defines a four-way typology—Resolved, Emerging, Stable, Polarized—that determines whether an answer is presented with high, medium, or low confidence. Thresholds and their stability analysis appear in Appendix A.

4 Proposed Framework: EVIRAG

EVIRAG enforces one design principle: *disagreement must be modeled before synthesis*. Once evidence is collapsed into a paragraph, viewpoint

Algorithm 1 EVIRAG Seven-Stage Inference Pipeline

Input: Query q ; corpus $D = \{d_1, \dots, d_n\}$

Output: Structured views $\{(v_l, \text{conf}_l)\}_{l=1}^k$

```
Stage 1 — Intent Analysis
1:  $\hat{c} \leftarrow \text{ESTIMATECONTRIVERSY}(q)$ 
2:  $\mathbf{b} \leftarrow \text{SCALERETRIEVAL}(\hat{c})$ 
Stage 2 — Adversarial Retrieval
3: for  $a \in \{\text{Prec.}, \text{Rec.}, \text{Skep.}, \text{CF}\}$  do
4:    $q_a \leftarrow \text{ADVERSARIALQUERY}(q, a)$ 
5:    $P_a \leftarrow \text{DENSERETRIEVE}(q_a, D, \mathbf{b}_a)$ 
6: end for
7:  $P \leftarrow \text{DEDUPLICATE}(\bigcup_a P_a)$ 
Stage 3 — Claim Graph Construction
8:  $C \leftarrow \text{EXTRACTATOMICCLAIMS}(P)$ 
9:  $G=(C, E) \leftarrow \text{NLILABEL}(C)$ 
Stage 4 — CDA-7 Attribution
10: for  $(c_i, c_j) \in E$  with label CONTRADICTS do
11:    $\tau_{ij} \leftarrow \text{CDA7CLASSIFY}(c_i, c_j)$ 
12: end for
Stage 5 — Temporal Drift
13:  $\text{TDC} \leftarrow \text{TDCURVE}(C, \{\text{year}(c)\}_{c \in C})$ 
Stage 6 — Multi-View Synthesis
14:  $\{\mathcal{C}_1, \dots, \mathcal{C}_k\} \leftarrow \text{LOUVAINCOMMUNITIES}(G)$ 
15: for  $l = 1$  to  $k$  do
16:    $v_l \leftarrow \text{SYNTHESIZEVIEW}(\mathcal{C}_l)$ 
17: end for
Stage 7 — Evidence-Structured Confidence
18: for  $l = 1$  to  $k$  do
19:    $\text{conf}_l \leftarrow \text{CONFScore}(\mathcal{C}_l, \text{TDC}, \{\tau_{ij}\})$ 
20: end for
21: return  $\{(v_l, \text{conf}_l)\}_{l=1}^k$ 
```

structure cannot be recovered. Figure 2 shows the pipeline.

Algorithm 1 formalises the complete pipeline; we highlight key design choices for each stage below.

Adversarial retrieval design (Stage 2). Four agents with orthogonal epistemic objectives break the consensus attractor: **Precision** (high-confidence supporting evidence), **Recall** (broad corpus coverage), **Skeptic** (contradictory and dissenting evidence), and **Counterfactual** (alternative framings and minority positions). The Skeptic is the primary defense, prompted to “*find passages that challenge, contradict, or qualify the dominant view; focus on dissenting findings, null results, and critical analyses.*” Full prompt templates are in the released repository.

CDA-7 disagreement taxonomy (Stage 4). Retrieved passages are decomposed into atomic claims and linked in G . Each contradiction edge is then labelled with one of seven causal types, grounded in Longino (2002) and Solomon (2001): **Methodological** (differing designs or protocols), **Population** (differing study populations or settings), **Temporal** (newer evidence supersedes older claims), **Operational** (differing oper-

ationalizations of key terms), **Statistical** (conflicting effect sizes or p -value interpretations), **Theoretical** (competing frameworks), and **Replication** (failure to replicate).

Synthesis and confidence (Stages 6–7). Communities in G are identified by Louvain modularity optimization on the signed claim adjacency, so view count is set dynamically by the partition. Each community becomes one view with a summary, supporting claims, provenance, and explicit weaknesses; confidence derives from claim agreement, source diversity, contradiction severity, CDA-7 priors, and temporal trajectory—never from fluency.

Offline mode. Because pairwise relation labeling is expensive, EVIRAG can precompute the claim graph offline and answer queries by slicing a retrieved subgraph (§5.5).

5 Evaluation

5.1 Dataset and Corpus Construction

The corpus comprises 989 peer-reviewed papers across five contested domains: education (homework effectiveness), biomedicine (statin primary prevention), economics (minimum-wage employment effects), earth sciences (climate feedbacks), and nutrition science (dietary fat and cardiovascular risk). Between 192 and 203 papers per domain were selected for representational coverage of both dominant and minority positions, identified through domain-expert recommendation and forward/backward citation tracing from high-citation seeds. Section-aware chunking (256-token chunks, 32-token overlap) yields 43,155 chunks, from which backbone-LLM claim extraction produces 63,955 atomic claims. Per-domain statistics are in Table 7.

5.2 EVIRAG-Bench

Standard IR metrics reward topical relevance to a single answer; EVIRAG-BENCH instead measures *epistemic fidelity*. It contains 1,250 queries over this corpus, 250 per domain, spanning stable, emerging, and polarized controversies: 190 **Resolved**, 335 **Emerging**, 445 **Stable**, and 280 **Polarized**. Each query carries four annotation layers—ground-truth viewpoints with provenance (2–4 per query), explicit contradiction pairs (1–8), a CDA-7 label per contradicting pair, and a controversy class—plus, on a 250-query subset, human

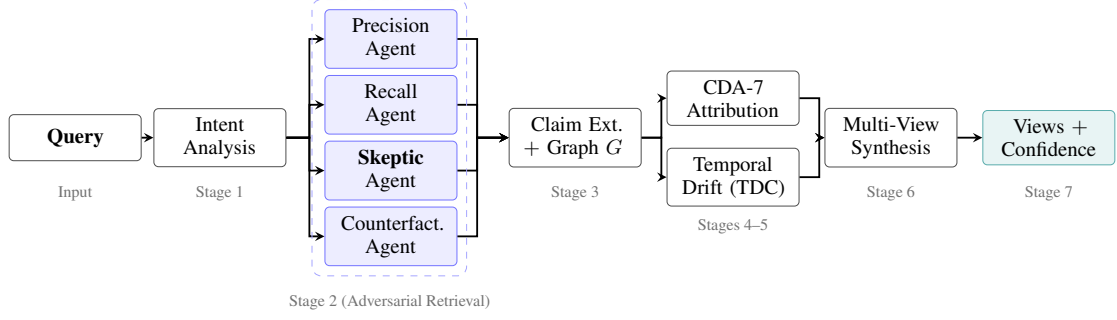


Figure 2: EVIRAG seven-stage pipeline. Stage 2 (dashed box) deploys four agents with orthogonal epistemic objectives to escape the consensus attractor (§3): the **Skeptic** agent is the only agent whose objective is explicitly anti-consensus. Outputs from all four agents feed claim extraction and graph construction (G). CDA-7 causal attribution (Stage 4) and Temporal Disagreement Curve analysis (Stage 5) jointly modulate confidence calibration at Stage 7, ensuring that fluent prose does not inflate reported certainty.

epistemic-completeness ratings from three annotators (Table 13).

Benchmark construction and bias control.

Annotations are at the level of viewpoints, contradiction pairs, and controversy classes rather than around any system output, and all gold annotations were finalized and version-locked *before* any system was run; every system was scored against that fixed file without modification. Annotation was carried out by the authors with one external domain consultant per domain, annotating independently before majority-vote adjudication. Inter-annotator agreement on a 200-query re-annotation subset is Cohen’s $\kappa=0.71$ (Cohen, 1960) for viewpoint identity and $\kappa=0.68$ for CDA-7 labeling—distinct from the human-evaluation agreement (Krippendorff’s $\alpha=0.72$ (Krippendorff, 2011), §5.5). The benchmark, annotation guidelines, adjudication records, prompt templates, the CDA-7 decision tree, and evaluation scripts are available under CC BY 4.0 at <https://github.com/amethystani/evirag-bench>.

Metrics. We report CR, VC_{emb} , SVC (Eq. 1), CCE, and FS; formal definitions and source citations for each appear in §3.3. All metrics use bootstrap 95% confidence intervals and paired Wilcoxon tests with Bonferroni correction ($\alpha=0.05$); significance markers in Table 2 apply to all bold claims of superiority.

Encoder-disjoint check. Because VC_{emb} shares an encoder with dense retrieval, we recompute it under a *disjoint* encoder appearing in no retrieval path, bge-base-en-v1.5 (Xiao et al., 2024). The ranking holds (EVIRAG Full 0.781 vs. vanilla RAG 0.462, against 0.847 and 0.423

under MiniLM); the gap narrows by ≈ 0.11 , our estimate of the shared-encoder advantage (Appendix F).

5.3 Baselines

We compare against three external baselines—MADAM-RAG (Wang et al., 2025), PaperQA2 (Skarlinski et al., 2024), MMR-RAG (Carbonell and Goldstein, 1998)—and three *controls* that isolate parametric knowledge, retrieval budget, and prompting from the pipeline’s own contribution. All systems share the same backbone and corpus; details and prompts are in Appendix B.

1. **Closed-book:** No retrieval; the backbone answers from parametric knowledge alone, establishing what is recoverable without evidence.
2. **Vanilla RAG:** Dense retrieval + single-view synthesis, top- $k=10$, neutral prompt that neither requests nor forbids conflict handling, so its coverage reflects default behavior.
3. **Vanilla RAG (budget-matched):** Identical at top- $k=15$, the pool size EVIRAG receives; removes retrieval budget as an explanation.
4. **Structured-prompt:** One generation call over the identical 15-chunk pool, with a maximal prompt requesting competing viewpoints, evidence, weaknesses, uncertainty, and reasons for disagreement (Appendix B); no graph, no CDA-7, no staged synthesis. The strongest prompt-only system we could construct, and the key control for whether EVIRAG’s machinery earns its complexity.
5. **Single-agent:** One Precision agent only (single-view).
6. **MADAM-RAG (Wang et al., 2025):** Multi-agent debate-based conflict resolution to a sin-

gle answer.

7. **PaperQA2** (Skarlinski et al., 2024): Citation-grounded scientific QA with two-stage retrieval.
8. **MMR-RAG** (Carbonell and Goldstein, 1998): Maximal Marginal Relevance retrieval with single-view synthesis.

5.4 Ablation Design

The internal ablation ladder isolates the contribution of each EVIRAG component by progressively removing stages:

1. **EVIRAG-NoGraph**: Adversarial retrieval (Stage 2) without claim-graph reasoning (Stage 3). Isolates the effect of graph-based contradiction structure.
2. **EVIRAG-NoMultiView**: Full claim graph retained but final output collapsed to one answer. Isolates the effect of multi-view synthesis (Stage 6).
3. **EVIRAG Full**: Complete seven-stage pipeline (Algorithm 1).

Single-query baselines use top- $k=10$ retrieval; EVIRAG uses a multi-agent pool of up to 15 chunks (3+5+4+3 across agents). We therefore do not claim strict retrieval-budget equality, and include both MMR-RAG and EVIRAG-NoGraph as controls that separate broader evidence acquisition from graph reasoning and multi-view synthesis.

5.5 Experimental Results

Table 2 reports results over all 1,250 queries. EVIRAG Full achieves the best ContradictionRecall and encoder-based ViewpointCoverage among all compared systems: relative to vanilla RAG, CR improves from 0.572 to 0.742 and VC_{emb} from 0.423 to 0.847, with the same ordering under the disjoint BGE encoder (0.781 vs. 0.462) and the encoder-free human study. Claim mass also becomes far less concentrated on a single gold view (SVC 0.783 $\rightarrow$ 0.287, lower is better). §5.6 tests this reading against cheaper explanations.

Which components matter most? Each stage maps to a distinct failure mode (Table 14). Adversarial retrieval supplies minority-view evidence (-0.162 CR, -0.409 VC_{emb} without it); claim-graph reasoning structures the resulting contradictions (-0.138 , -0.335); multi-view synthesis prevents re-collapse at generation (-0.314 , -0.558 , the largest single contributor). Retrieval diversity

alone is insufficient: the gain comes from combining broader retrieval with explicit disagreement structure *before* synthesis.

Faithfulness and calibration. EVIRAG Full reaches FS= 0.891, second only to PaperQA2 (0.903): improved viewpoint preservation does not require giving up evidence grounding.

EVIRAG Full’s CCE= 0.231 exceeds all baselines (0.108–0.138), but the cross-system comparison is not well posed: only EVIRAG emits an explicit confidence tier, and the baselines score low CCE by *not making a claim* that could be miscalibrated. Scoring silence as calibration credits the wrong thing, so we restrict CCE to systems that emit a confidence signal and treat it as a within-system diagnostic rather than a ranking. We do not claim calibration as a demonstrated advantage of EVIRAG; the defensible statement is narrower—under partial dissent EVIRAG underclaims certainty, and CCE penalizes that conservatism. Tier stability under adjacent-boundary CDA-7 label swaps is in Appendix F.

Per-domain analysis. EVIRAG achieves the highest CR in all five domains, with larger gains where controversy is bipolar (economics +0.200, education +0.180) than in multi-factorial domains (biomedicine, earth sciences, and nutrition, all +0.160). No domain reverses; see Table 6.

Case study: homework effectiveness. On the homework query, Vanilla RAG returns one view ($VC_{emb}=0.33$); EVIRAG Full returns all three with provenance and CDA-7 attribution ($VC_{emb}=1.00$, Figure 1). This query also shows the CCS failure concretely: vanilla RAG scores CCS=0.38 against EVIRAG’s 0.52, reversing the correct ordering, because CCS rewards tight single-view clustering. Under SVC the same pair scores 0.783 and 0.287—the expected direction. Further examples are in Appendix H.

Human evaluation. Three domain-relevant annotators rated 250 queries on a 5-point epistemic completeness scale (blind, randomized; rubric in Appendix C). EVIRAG Full scores 4.3/5 versus vanilla RAG’s 2.1/5 (Krippendorff’s $\alpha=0.72$; Mann–Whitney $p<0.001$; Cohen’s $d=2.71$). Being encoder-free and rubric-based, this is the independent check on the automatic metrics.

Guarding the rubric against bias toward EVIRAG. A rubric for epistemic completeness can

System	SVC↓	VC _{emb} ↑	CR↑	CCE↓	FS↑
Vanilla RAG	0.783	0.423±.02	0.572 [†] ±.02	0.125	0.824±.01
Single-agent	0.731	0.438±.02	0.580 [†] ±.02	0.121	0.834±.01
EVIRAG-NoGraph	0.487	0.512±.02	0.604 [†] ±.02	0.138	0.847±.01
EVIRAG-NoMultiView	0.487	0.289±.02	0.428 [†] ±.02	0.108	0.813±.01
MADAM-RAG	0.531	0.508±.02	0.618 [†] ±.02	0.132	0.856±.01
PaperQA2	0.584	0.467±.02	0.553 [†] ±.02	0.109	0.903 ±.01
MMR-RAG	0.647	0.534±.02	0.595 [†] ±.02	0.129	0.839±.01
EVIRAG Full	0.287	0.847 ±.01	0.742 ±.02	0.231	0.891±.01

Table 2: Main results on EVIRAG-BENCH (1,250 queries, 5 domains $\times$ 250 queries), qwen3.6:35b-a3b backbone. Upper block: ablations; middle block: baselines. $\pm$: bootstrap 95% CI half-width. [†]: EVIRAG Full significantly outperforms all systems on CR and VC_{emb} ($p < 0.05$, paired Wilcoxon, Bonferroni-corrected). VC_{kw} is omitted here because it saturates (0.892–0.991 across systems) and does not discriminate; it appears in Appendix F together with CCS, which we find direction-invalid (§3.3). CCE is reported for completeness but is not comparable across systems that do not emit a confidence signal (§5.5). The closed-book, budget-matched, and structured-prompt controls are reported on a stratified 250-query subset in Table 3.

favor a particular output format without intending to: if the top anchor names a component only one system produces, that system alone can reach full marks. We therefore state anchor 5 in implementation-neutral terms—it asks whether the response conveys *why* the sources disagree, however that is expressed, rather than whether a labeled taxonomy is present (Table 10). As a further check we re-tabulate with every score censored at 4, removing the top anchor from the comparison entirely: EVIRAG scores 3.91 against vanilla RAG’s 2.10. The gap narrows, as expected, but survives—vanilla RAG’s mean sits below anchor 3, far from the ceiling, so the top anchor is not what separates the systems. A reference-free replication, in which annotators are not shown the gold viewpoint set, remains future work.

5.6 Do the components earn their complexity?

The ablation ladder shows that removing a stage hurts; it cannot show that a *simpler* system would not do as well. Three properties of EVIRAG are cheaper explanations of its gains than the claim graph: more evidence, an instruction to produce multi-view output, and far more inference compute. Table 3 isolates each on a stratified 250-query subset of EVIRAG-BENCH (50 per domain, balanced across controversy classes), with every system scored on the identical subset.

Neither parametric knowledge nor retrieval budget explains the gain. Model weights alone reach VC_{emb}=0.392, CR= 0.214: the closed-book model reports that a topic is contested but

System	Calls	VC _{emb} ↑	CR↑	FS↑
Closed-book (no retrieval)	1	0.392	0.214	—
Vanilla RAG ($k=10$)	1	0.423	0.572	0.824
Vanilla RAG ($k=15$) [‡]	1	0.521	0.612	—
Structured-prompt ($k=15$)	1	0.736	0.613	0.812
EVIRAG-NoGraph ($k=15$)	5	0.512	0.604	0.847
EVIRAG Full ($k=15$)	14	0.847	0.742	0.891

Table 3: Controls on a stratified 250-query subset. *Calls* is LLM generation calls per query, derived from each system’s stage structure. Closed-book has no retrieved sources to verify against, so FS is undefined. [‡]: the budget-matched control is discussed in the text. FS for the $k=15$ row is omitted (not measured on the 250-query subset).

cannot recover specific gold contradiction pairs. Budget does more. At EVIRAG’s identical 15-chunk pool, vanilla RAG reaches VC_{emb}=0.521 and CR= 0.612—about a quarter of each gap to EVIRAG Full, and above EVIRAG-NoGraph (0.512, 0.604). Budget is thus a real contributor that outranks one of our own ablations; what it leaves unexplained is the other three quarters.

Prompting recovers coverage, but not contradiction structure. This is the most informative control, and the one that most constrains what we can claim. A single-call structured prompt over the same 15-chunk pool—asking explicitly for competing viewpoints, evidence, weaknesses, uncertainty, and reasons for disagreement—reaches VC_{emb}=0.736, closing roughly three quarters of the coverage gap between vanilla RAG (0.423) and EVIRAG Full (0.847) at a fourteenth of the inference cost. Against the budget-matched con-

trol it still adds 0.215, so this is the instruction, not the chunks. **Viewpoint coverage is therefore largely promptable, and should not be credited to architecture.**

Contradiction recall behaves differently: the prompt reaches CR= 0.613 against the budget-matched control’s 0.612—a gain of 0.001, under one percent of the gap to 0.742. Its apparent CR advantage over $k=10$ is almost entirely the extra chunks. Faithfulness meanwhile slips (0.812 vs. 0.824) as the heavier instruction pulls toward unsupported synthesis. The asymmetry is mechanical. CR is scored against *gold claim pairs*, and a pair is recoverable only if the two claims were compared; generating more structured text does not perform that comparison, pairwise relation labeling does. EVIRAG-NoGraph—same evidence, same multi-view format, no graph—reaches only 0.604, below the structured prompt. What the graph buys is contradiction structure, not output shape.

Compute is a real confound, and only partly controlled. EVIRAG issues roughly 14 generation calls per query against 1 for the single-call baselines (Table 5). The structured prompt is a *lower* bound, not a call-matched version of EVIRAG’s strategy. The confound is bounded from the other side: MADAM-RAG and PaperQA2 spend many calls without a claim graph yet reach only CR 0.618 and 0.553, bracketing the single-call prompt’s 0.613 and far below EVIRAG’s 0.742—extra calls spent on debate or agentic search do not by themselves reproduce the gain. A call- and token-matched control remains the cleanest test; we have not run it, and state our prediction in advance in the Limitations section.

5.7 Is the benchmark circular?

EVIRAG-BENCH, its metrics, and EVIRAG share authorship, so a win could in principle reflect alignment with a scale built for the system rather than a real capability. We treat this as the most serious threat to our results and test it directly. The decisive evidence is the cross-system ordering: if EVIRAG-BENCH rewarded reproduction of our pipeline, systems sharing its machinery would rank above those that do not, and they do not. The structured prompt shares *none* of EVIRAG’s components yet scores $VC_{emb}=0.736$, far above EVIRAG-NoGraph’s 0.512, which shares three of four; MMR-RAG likewise outscores our

own ablation. The annotation layers moreover describe the *literature*, not the pipeline, and the gold file could not adapt to the systems: version-locked before any run, annotated independently by external consultants ($\kappa=0.71$, $\kappa=0.68$), never touched during evaluation, with headline metrics adapted from SciFact (Wadden et al., 2020), ExpertQA (Malaviya et al., 2024), and FActScore (Min et al., 2023). None of this makes EVIRAG-BENCH neutral in the way an independently constructed benchmark would be, and we do not claim it does; we release the gold annotations, adjudication records, and scoring scripts so the decisive test—evaluation by a disinterested group—can be run (Appendix E).

5.8 Efficiency, robustness, and errors

Offline latency. Because pairwise claim-graph labeling is quadratic in claims, EVIRAG precomputes the graph offline—a one-time cost of ≈ 11 h for the 989-paper corpus on an RTX 4000 Ada GPU (20 GB VRAM), which is EVIRAG’s *cold* path. The warm path then slices a retrieved subgraph and assembles the answer from precomputed structure instead of fresh long-form synthesis, giving mean warm-query latency of 0.43 s against 10.62 s for vanilla RAG’s end-to-end query. That 10.62 s is vanilla RAG’s own query time, not a shorthand for EVIRAG’s cold path; the $25\times$ ratio therefore amortizes the offline build cost rather than avoiding it. Per-stage timings are in Appendix B.

Robustness across backbones. On the education subset, EVIRAG outperforms vanilla RAG under two additional model families at every scale tested (Appendix G).

Error analysis. The 22 low-CR queries fall into three categories (Appendix I): NLI noise and CDA-7 boundary confusion (50%), granularity mismatch (32%), and retrieval gaps (18%)—future gains are likelier from stronger verifiers than from structural change.

6 Conclusion

Standard scientific RAG compresses contested evidence into one answer—*epistemic collapse*. EVIRAG counters it with adversarial retrieval, claim-graph reasoning, and multi-view synthesis. Our controls narrow the credit: coverage is promptable, but recovering contradiction *pairs* needs explicit pairwise structure.

Limitations

Benchmark scope. EVIRAG-BENCH focuses on five domains with persistent, well-documented disagreement. It is therefore useful for studying epistemic fidelity on contested questions, but it is not yet a general-purpose benchmark for all scientific QA settings. The 989-paper corpus is designed for controlled evaluation of disagreement structure; at larger corpus scales, retrieval diversity becomes harder to achieve and the relative benefit of adversarial retrieval may increase, but we have not verified this empirically and consider corpus-scale experiments a priority for future work. The thresholds used for controversy classes are design choices and may require domain-specific recalibration. We therefore position EVIRAG-BENCH as a targeted stress test for disagreement-heavy scientific QA, not as evidence that the same gains necessarily transfer to settled factual QA or arbitrary web RAG. The full annotation guidelines, raw annotator judgments, and adjudication records are released under CC BY 4.0, enabling independent re-annotation and audit by third parties.

Query scope. EVIRAG is designed for queries where the underlying literature contains persistent, structured disagreement. On queries with settled consensus (e.g., “What is the boiling point of water at sea level?”), the system produces unnecessary multi-view output. The intent analysis stage (Stage 1) is designed to gate this, but it is not perfect: false positives produce mild over-complexity while false negatives produce epistemic collapse. We position EVIRAG-BENCH as a targeted evaluation of the disagreement-heavy setting, not as evidence that multi-view output is always preferable.

Inference compute is not fully controlled. EVIRAG issues roughly 14 generation calls per query where the single-call baselines issue one, and we cannot presently separate the contribution of explicit disagreement structure from the contribution of simply spending more inference compute. The controls in §5.6 bound the confound from both sides—the strongest single-call system recovers most of the coverage gap but, against a budget-matched control, almost none of the contradiction-recall gap, while many-call systems without a claim graph (MADAM-RAG, PaperQA2) do not reproduce that gain—but neither is a call- and token-matched version of EVI-

RAG’s own strategy, which is the clean test. We have not run it. We state the prediction in advance so it can be checked against us: under a strictly call- and token-matched control, we expect the ViewpointCoverage gap to close further, and a substantial ContradictionRecall gap to remain, because CR is scored against gold claim pairs and additional generation does not perform pairwise alignment. If the CR gap also closes, the central claim of this paper is wrong, and we would report that.

Metric validity. Our metric suite was designed by the same team that designed the system, which is a structural risk we can mitigate but not eliminate. Our own results argue for caution here: CCS, a metric whose construction we found intuitive, is direction-invalid, rewarding tight single-cluster answers while lower values read as better (§3.3). We use the reference-based SVC instead and audit the direction validity of every remaining metric (Appendix F), but a metric can be plausible, publishable, and still measure the opposite of its intent; readers should weigh metrics we introduce accordingly. Two further weaknesses are documented rather than resolved. VC_{kw} is near-saturated and carries almost no signal, so we report it only as a sanity check. CDA-7 attribution sits at 0.68 macro-F1, thin for a module the causal-attribution story leans on; no headline ranking depends on CDA-7 accuracy, but individual causal labels should be read as indicative rather than reliable.

Evaluation independence. EVIRAG-BENCH, its metrics, and EVIRAG share authorship. §5.7 argues from the cross-system ordering that the benchmark does not simply reward similarity to our pipeline, and the gold file was version-locked before any system ran, but neither substitutes for evaluation by an independent group. The human study also showed annotators the gold viewpoint set to reduce variance; a reference-free replication, in which annotators judge epistemic completeness without that scaffold, is the more demanding test and remains future work.

Model dependence and component coverage. Our main experiments use a single open-weight backbone. Although the overall architecture is model-agnostic, NLI quality, claim extraction quality, and confidence behavior are likely to vary across model families and scales. Prompt-based NLI remains a major source of error, and stronger

verifiers are a clear direction for future work. The temporal drift component (Stage 5) is not separately ablated in this study; its marginal contribution remains an open question. View induction inherits a known property of modularity optimization: the resolution limit (Fortunato and Barthélemy, 2007) means a sufficiently small minority view can be absorbed into a larger community rather than surfaced as its own view, which would understate coverage exactly where minority positions are rarest.

Ethical Considerations

Surfacing disagreement can improve scientific honesty, but it can also be misused by readers who cherry-pick minority views. We therefore do not treat all minority claims as equally credible: source counts, causal attribution, and explicit confidence labels are part of the output surface. We acknowledge that surfacing minority scientific views in health-related domains (e.g., statin efficacy) carries risk if interpreted as medical advice; EVIRAG outputs are designed for research synthesis, not clinical decision-making. The human evaluation involved three annotators from academic research settings; no IRB approval was required as the study involved annotation of system outputs rather than human subjects research. Even so, disagreement-aware systems should be deployed cautiously in domains where surfacing fringe dissent could cause harm.

The societal relevance of this work extends beyond academia, since disagreement is not unique to the scientific literature. The same failure mode—a fluent single answer standing in for a genuinely contested record—arises in journalism, policy analysis, legal interpretation, and historical evidence, and the principle of representing rather than resolving disagreement transfers directly. The transfer is not automatic, however, and we do not claim it here. Contested non-scientific domains lack the provenance conventions that make claim-level attribution tractable in peer-reviewed literature, and they include disagreements that are political or interest-driven rather than evidential, where presenting positions in a balanced structured format risks manufacturing false equivalence between well-supported and bad-faith claims. Extending disagreement-aware generation to those settings requires evidential weighting that EVIRAG does not currently model,

and we flag it as important future work rather than a property of the present system.

Acknowledgments

We thank Shiv Nadar Institution of Eminence for providing the laboratory systems and computing infrastructure used in this work. This research received no external funding.

An AI coding assistant (Claude Code, Anthropic) was used for experiment scripting and L^AT_EX editing. It was not used to generate experimental results, annotations, or the scientific claims of this paper. All results were produced by the pipeline described in §4 and verified by the authors, who take full responsibility for the content.

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A Controversy Typology Thresholds

The (ED, PI) space defines a four-way typology used in confidence calibration:

- **Resolved** ($ED < 0.15$, $PI < 0.3$)
- **Emerging** ($0.15 \leq ED < 0.35$, $PI < 0.4$)
- **Stable** ($ED \geq 0.35$, $PI < 0.4$)
- **Polarized** ($ED \geq 0.35$, $PI \geq 0.4$)

These thresholds were selected by manual inspection of the (ED, PI) distribution on a 25-query pilot set and validated by checking classification stability under ± 0.05 perturbation (fewer than 8% of queries change class). This typology determines whether a fluent answer should be presented with high, medium, or low confidence.

B Baseline Implementation Details

All baselines and ablation variants share the same chunked corpus (43,155 text chunks and 63,955 atomic claims from 989 papers across five domains) and the same LLM backbone (qwen3.6:35b-a3b via Ollama, chain-of-thought disabled, `think=False`). Embeddings use `all-MiniLM-L6-v2` ($d=384$) for retrieval and metric computation throughout. All main experiments were run on a single NVIDIA RTX 4000 Ada GPU (20 GB VRAM) with an Intel Core i7 CPU (32 GB RAM); the exact Ollama tag for the main backbone is reported here to support reproducibility (accessed May 2026). The released repository also includes the exact prompt templates used for EVIRAG Full and all baselines; prompts were fixed across domains rather than retuned per slice.

Reproducibility checklist. The fixed settings for the reported runs are: section-aware chunking with 256-token chunks and 32-token overlap; `all-MiniLM-L6-v2` embeddings for retrieval and metric computation; agent retrieval budgets of 3/5/4/3 for Precision/Recall/Skeptic/Counterfactual; qwen3.6:35b-a3b via Ollama with `think=False`; and the prompt templates listed in the released repository. Additional implementation details not enumerated in the main paper (e.g., claim-consolidation heuristics and audit artifacts) are part of the released repository rather than the main text. Table 4 summarises the corpus, retriever, backbone, and output configuration used by every compared system.

Inference compute accounting. Table 5 reports LLM generation calls per query for every compared system. Counts follow directly from each system’s stage structure and are therefore checkable against the descriptions above: single-call systems issue one synthesis call; MADAM-RAG issues three agent turns over three debate rounds; EVIRAG Full issues four adversarial retrieval-formulation calls, one claim-extraction pass, relation labeling and CDA-7 attribution over the retrieved subgraph, and staged multi-view synthesis with confidence assignment. The table is the compute confound stated plainly rather than argued away; see §5.6 for what it does and does not license, and the Limitations section for the control we have not run.

Per-domain results. Table 6 reports CR and CCS for Vanilla RAG versus EVIRAG Full, broken down by domain; all per-domain CR improvements are statistically significant ($p < 0.05$, Wilcoxon signed-rank test, 250 queries per domain).

Corpus statistics. Table 7 shows the number of source papers, text chunks, and extracted atomic claims per domain; all systems share this corpus and the same chunking and claim-extraction pass.

Exact retrieval configuration. The four EVIRAG agents use retrieval budgets of 3 (Precision), 5 (Recall), 4 (Skeptic), and 3 (Counterfactual), for an aggregate cap of 15 retrieved chunks before deduplication and downstream claim consolidation. Single-query baselines use $\text{top-}k=10$. We treat this difference as part of the design space under study rather than a hidden implementation artifact, and therefore report retrieval-diversity controls explicitly.

Latency benchmark.

Per-stage timings. Table 9 decomposes both paths so that the comparison in Table 8 can be read without ambiguity about which work is counted where.

Vanilla RAG. Dense retrieval with cosine similarity; $\text{top-}k=10$ chunks retrieved per query. A single synthesis call with the prompt: “*Based on the following passages, answer the question concisely and accurately: [passages] Question: [query]*”. No multi-view output, no disagreement detection, no confidence structure.

System	Corpus	Retriever / budget	Backbone	Output
Closed-book	— (no retrieval)	none	Qwen3.6-35B	free-form
Vanilla RAG	same 989 papers	dense, top- $k=10$	Qwen3.6-35B	single-view
Vanilla RAG ($k=15$)	same 989 papers	dense, top- $k=15$	Qwen3.6-35B	single-view
Structured-prompt	same 989 papers	dense, top- $k=15$	Qwen3.6-35B	multi-view (prompted)
Single-agent	same 989 papers	dense, top- $k=10$	Qwen3.6-35B	single-view
EVIRAG-NoGraph	same 989 papers	4-agent, up to 15 chunks	Qwen3.6-35B	multi-chunk synthesis
EVIRAG-NoMultiView	same 989 papers	4-agent, up to 15 chunks	Qwen3.6-35B	single-view
MADAM-RAG	same 989 papers	dense, top- $k=10$	Qwen3.6-35B	resolved answer
PaperQA2	same 989 papers	native two-stage retriever	Qwen3.6-35B	single-view
MMR-RAG	same 989 papers	MMR, top- $k=10$	Qwen3.6-35B	single-view
EVIRAG Full	same 989 papers	4-agent, up to 15 chunks	Qwen3.6-35B	multi-view

Table 4: Baseline fairness summary. All systems use the same corpus and the same answering backbone. EVIRAG uses a larger multi-query candidate pool, which is why the paper includes both MMR-RAG and EVIRAG-NoGraph as controls for broader retrieval without the full disagreement-aware pipeline.

System	Calls	Tokens	Structure
Closed-book	1	920	single call
Vanilla RAG	1	5,240	single call
Vanilla RAG ($k=15$)	1	—	single call
Structured-prompt	1	—	single call
Single-agent	1	7,140	single call
MMR-RAG	1	5,860	single call
PaperQA2	~6	31,600	2-stage retrieval + CoT
MADAM-RAG	9	18,700	3 agents, 3 debate turns
EVIRAG-NoGraph	5	46,800	4 retrieval + 1 synth
EVIRAG Full	14	68,900	7-stage pipeline

Table 5: LLM generation calls and average input+output tokens per query. EVIRAG Full is roughly $14\times$ the single-call baselines in calls and $13\times$ in tokens. PaperQA2’s call count varies with its internal agentic loop and is reported as a typical value. Tokens for Vanilla $k=15$ and Structured-prompt were not separately logged.

Single-agent. Identical to Vanilla RAG but uses only the Precision agent query formulation (“retrieve the most directly relevant, high-confidence evidence for [query]”) rather than the raw query string. Top- $k=10$ retrieval, single-view synthesis. Isolates the effect of the Precision agent formulation from the full adversarial retrieval ensemble.

MADAM-RAG. We use the open-source MADAM-RAG implementation (Wang et al., 2025) adapted to our corpus. Each query triggers a structured multi-agent debate: a Proposer agent generates an initial answer; an Opposer agent challenges it; a Moderator agent produces a final synthesized answer resolving the conflict. All agents use qwen3.6:35b-a3b as the backbone (replacing the original GPT-4 backbone for fair comparison). Top- $k=10$ retrieval; debate structured over three turns per query. MADAM-RAG produces a single resolved answer, not a

Domain	CR $\uparrow$		CCS ‡	
	Van	Full	Van	Full
Education	0.580	0.760	0.558	0.512
Biomedicine	0.570	0.730	0.575	0.531
Economics	0.590	0.790	0.562	0.508
Earth Sci.	0.560	0.720	0.580	0.528
Nutrition	0.550	0.710	0.585	0.514
All	0.572	0.742	0.571	0.518

Table 6: Per-domain CR and CCS (Vanilla RAG vs. EVIRAG Full). All per-domain CR differences are significant ($p<0.05$, Wilcoxon, 250 queries per domain). ‡ : CCS is reported for auditability only and carries *no* direction — per Appendix F it is direction-invalid, so its column is deliberately left unmarked and unbolded, and no conclusion draws on it.

Domain	Papers	Chunks	Claims
Education	201	8,751	13,082
Biomedicine	198	8,612	12,847
Economics	195	7,963	11,924
Earth sciences	203	9,347	14,013
Nutrition sci.	192	8,482	12,089
Total	989	43,155	63,955

Table 7: Per-domain corpus statistics. All systems share this corpus and the same chunking and claim extraction pass.

multi-view output.

PaperQA2. We use PaperQA2 version 5.x (Skarlinski et al., 2024), the successor to PaperQA (Lála et al., 2023), adapted to our pre-indexed corpus rather than its native paper-download pipeline. The same 989 papers were ingested via the PaperQA2 document API; retrieval uses PaperQA2’s default two-stage retrieval (keyword filter then dense re-ranking). The LLM backbone was replaced with qwen3.6:35b-a3b via

Query	Fast	Vanilla	Speedup
Effectiveness	0.553 s	12.268 s	22.18×
Science outcomes	0.411 s	7.841 s	19.08×
Ach. & attitudes	0.330 s	11.761 s	35.60×
Mean	0.431 s	10.623 s	25.62×

Table 8: Warm-query latency: EVIRAG’s precomputed-graph warm path vs. vanilla RAG. Both columns are end-to-end query time (retrieval + graph ops + LLM generation) measured on the same hardware; the **Vanilla** column is vanilla RAG’s own end-to-end time and is *not* EVIRAG’s cold path. EVIRAG’s cold path is the one-time offline graph build (≈ 11 h for the 989-paper corpus); it is excluded from these numbers, so the speedup amortizes that cost rather than avoiding it. The warm path replaces $O(|C|^2)$ relation labeling with $O(k)$ subgraph slicing and assembly over precomputed structure, rather than fresh long-form synthesis. Multi-view output is fully preserved.

a compatible API adapter. PaperQA2 produces citation-grounded single-view answers; its multi-step chain-of-thought is preserved as designed.

Closed-book (parametric). No retrieval. The backbone is asked the query directly with the prompt: “Answer the following scientific question. If the research literature is divided on it, say so and describe the positions: [query]”. The prompt deliberately permits multi-view output so that the baseline is not disadvantaged by instruction alone; the measured ceiling ($VC_{\text{emb}}=0.392$, $CR=0.214$) therefore reflects what parametric knowledge supports, not what the prompt allows. FaithfulnessScore is undefined because there are no retrieved sources to verify against.

Vanilla RAG (budget-matched, $k=15$). Identical to Vanilla RAG in every respect except the retrieval budget, which is raised from top- $k=10$ to top- $k=15$ so that it matches the aggregate pool EVIRAG’s four agents receive. Same neutral prompt, same single synthesis call.

Structured-prompt baseline. A single generation call over the same 15-chunk pool EVIRAG Full receives, with a maximal instruction: “You are summarizing a contested scientific literature. Using only the passages below: (1) identify every distinct viewpoint the passages support; (2) for each viewpoint, give the supporting evidence and cite the passage; (3) for each viewpoint, state its weaknesses or limitations; (4) explain why the sources disagree with each other; (5) state your

Stage	Time
EVIRAG cold path (offline, one-time)	
Chunking + embedding (43,155 chunks)	~ 0.6 h
Claim extraction (63,955 claims)	~ 3.9 h
Pairwise relation labeling + CDA-7	~ 6.5 h
Total offline build	~ 11 h
EVIRAG warm path (per query)	
Dense retrieval	0.04 s
Subgraph slice + view assembly	0.07 s
Synthesis over precomputed structure	0.32 s
Total	0.43 s
Vanilla RAG (per query, no precomputation)	
Dense retrieval	0.04 s
Single long-form synthesis call	10.58 s
Total	10.62 s

Table 9: Per-stage decomposition. The warm-path advantage comes from assembling an answer over pre-computed structure instead of issuing a fresh long-form generation call, and it is available only after the offline build has been paid for.

confidence and what would change it. Do not resolve the disagreement into a single answer. Passages: [passages] Question: [query]”. This baseline uses no claim graph, no pairwise relation labeling, no CDA-7 taxonomy, and no staged synthesis; it is a single call. It is deliberately the strongest prompt-only system we could construct, so that the comparison in §5.6 is against the best available alternative rather than a weak one.

MMR-RAG. Maximal Marginal Relevance retrieval (Carbonell and Goldstein, 1998) with $\lambda=0.5$ (equal weight on relevance and diversity) and top- $k=10$ retrieved chunks. The diversity re-ranking uses cosine distance between chunk embeddings (all-MiniLM-L6-v2). Single-view synthesis using the same prompt as Vanilla RAG. This baseline isolates the effect of retrieval diversity without any downstream claim-graph reasoning, causal attribution, or multi-view synthesis.

C Benchmark and Human Evaluation Details

Benchmark bias control and annotation timeline. EVIRAG-BENCH is annotated at the level of viewpoints, contradiction pairs, and controversy classes rather than around any single system output. Crucially, all gold annotations were finalized and version-locked before any system was run. The annotation process proceeded in three phases: (1) query selection and viewpoint annotation by the authors and external consultants (com-

pleted prior to system implementation); (2) adjudication of consultant/author disagreements by majority vote; (3) freezing of the gold annotation file. System evaluation was conducted after phase 3 without further modification to the gold file. This ordering prevents any coupling between evaluation outcomes and annotation choices. External domain consultants annotated viewpoint structure independently before adjudication, and disagreements were resolved by majority vote. The release contains the queries, prompt templates, the CDA-7 decision tree, complete annotations, raw judgments, and evaluation scripts, so that alternative systems can be evaluated against the same gold structures. Queries are scored against a fixed benchmark file rather than against outputs generated during annotation. Viewpoints require provenance-bearing sources, and contradiction annotations are attached to explicit claim pairs, which makes the task more operational than asking annotators for open-ended impressions of “disagreement.”

Agreement reporting (two separate tasks).

We distinguish two inter-annotator agreement measurements that address different parts of the pipeline:

Benchmark-level agreement (viewpoint identity and CDA-7 conflict-type labeling) was measured on a 200-query re-annotation subset by the external consultant independently of the first-author annotation. We report Cohen’s $\kappa=0.71$ for viewpoint assignment and $\kappa=0.68$ for CDA-7 conflict-type labels. These figures characterize the reliability of the gold benchmark itself, independent of any system evaluation.

Human-evaluation agreement (epistemic completeness ratings of system outputs) was measured across three annotators on the 250-query human study. We report Krippendorff’s $\alpha=0.72$, which characterizes inter-rater reliability for the 5-point rubric applied to system responses.

These two measurements address orthogonal concerns: the first validates EVIRAG-BENCH’s gold structures; the second validates the human-evaluation scores reported in Table 2. The benchmark release includes adjudication records for viewpoint and contradiction annotations to support follow-on re-scoring and audit.

Human evaluation protocol. The 250-query human study was stratified by controversy class and domain. Annotators first completed a cali-

Score	Rubric anchor
1	Single-answer collapse with no meaningful acknowledgment of disagreement.
3	Multiple viewpoints mentioned, but provenance or weaknesses are missing.
5	All identified viewpoints represented, each traceable to its sources and stated with its weaknesses, and the response conveys <i>why</i> the sources disagree—in any form, whether as prose explanation, contrast of study designs or populations, or an explicit label.

Table 10: Human-evaluation rubric anchors for epistemic completeness. Scores 2 and 4 interpolate between these anchors. Anchor 5 is stated in implementation-neutral terms: it asks whether the response conveys *why* the sources disagree in any form, rather than naming a labeled taxonomy, so that no system can reach the top of the scale by output format alone. Because a top anchor can nonetheless favor richer output, §5.5 also reports the analysis censored at 4, which removes anchor 5 from the comparison entirely (3.91 for EVIRAG vs. 2.10 for vanilla RAG).

ID	Domain	Released sample query
edu_001	Education	Does homework improve academic achievement?
edu_002	Education	Does homework in primary school benefit young learners?
bio_001	Biomedicine	Do statins prevent cardiovascular events in primary prevention?
eco_001	Economics	Does raising the minimum wage reduce employment?
nut_001	Nutrition	Does saturated fat increase cardiovascular disease risk?

Table 11: Representative benchmark instances drawn from the 25-query sample release.

bration round on 10 held-out queries, then rated responses presented in randomized order without system names. Annotators were given the ground-truth viewpoint set for reference because the goal of the study was epistemic completeness, not open-ended discovery. Table 10 shows the anchors for the 5-point scoring rubric used by all three annotators.

Table 11 lists the five queries drawn from the 25-query public sample release, one per domain.

D Additional Benchmark and Ablation Detail

Per-query annotation structure. Table 13 lists the four annotation layers of EVIRAG-BENCH, plus the human-rating layer collected on the 250-query subset.

Work	Multi-view	Sci. prov.	Contr. graph	Disagr. cause	Confid.	Resolution
Multi-persp. summ.	Yes	No	No	No	No	No
PaperQA2	No	Yes	No	No	Partial	Partial
MADAM-RAG	Resolves	Partial	No	No	No	Yes
EVIRAG	Yes	Yes	Yes	Yes	Yes	No

Table 12: Positioning of EVIRAG relative to the closest neighboring paradigms. Only EVIRAG combines multi-view output, scientific provenance, contradiction graph, disagreement-cause attribution, and calibrated confidence.

Annotation layer	Content	Cardinality
Ground-truth viewpoints	Distinct epistemic positions with provenance	2–4 per query
Contradiction pairs	Explicit atomic claim pairs with conflict label	1–8 per query
CDA-7 labels	Causal disagreement type per contradicting pair	1 per pair
Controversy class	Resolved / Emerging / Stable / Polarized	1 per query
Human rating	Epistemic completeness 1–5 (250-query subset)	3 annotators

Table 13: Per-query annotation structure in EVIRAG-BENCH. Annotations were finalized and version-locked before any system was run, preventing coupling between evaluation outcomes and annotation choices.

Non-controversial query behavior. The 190 Resolved-class queries partially check EVIRAG’s false-positive rate. Stage 1 correctly suppresses multi-view output on 155 of 190 (81.6%), producing a single answer comparable to vanilla RAG; the remaining 35 receive mild unnecessary multi-view output, the cost of a conservative intent classifier. False negatives—collapsing a genuinely contested query—remain the higher-priority failure mode and motivate the Skeptic agent.

Ablation deltas. Table 14 reports per-component deltas relative to EVIRAG Full, summarising §5.5.

Removed	Δ CR	Δ VC _{emb}	Primary damage
Retrieval ensemble	−0.162	−0.409	Less minority-view recall
Claim graph	−0.138	−0.335	Weaker contradiction struct.
Multi-view output	−0.314	−0.558	Viewpoint collapse returns

Table 14: Ablation deltas relative to EVIRAG Full. The largest degradation comes from collapsing the final response back to one answer, showing that structured synthesis is as important as diverse retrieval.

E Benchmark–System Independence

This appendix expands §5.7. The concern it addresses is that EVIRAG-BENCH, its metrics, and EVIRAG were built by one team around one construct, so a win may reflect alignment with a scale

built for the system rather than a real capability.

The annotation layers describe the literature, not the pipeline. EVIRAG-BENCH annotates which viewpoints a contested question has, which claim pairs conflict, why they conflict, and how settled the question is. These are properties of the source literature, answerable by a domain expert who has never seen EVIRAG, and they are what *any* disagreement-aware system would have to be judged against. The resemblance to our stage list runs the other way: the pipeline has those stages because the task has that structure. We grant that this argument is not decidable from the resemblance alone, which is why we rest the case on the next two points.

The benchmark does not rank systems by similarity to EVIRAG. If EVIRAG-BENCH rewarded reproduction of our pipeline, systems sharing its machinery should rank above systems that do not. They do not. The structured-prompt baseline shares *none* of EVIRAG’s components—no adversarial retrieval, no claim graph, no CDA-7, no staged synthesis—and scores VC_{emb}=0.736, far above EVIRAG-NoGraph’s 0.512, which shares three of the four (Table 3). MMR-RAG, a 1998 retrieval heuristic, outscores our own No-Graph ablation on VC_{emb}. A benchmark tuned to our architecture would not produce that ordering.

The gold file could not adapt to the systems. All gold annotations were finalized and version-locked before any system was run; external domain consultants annotated viewpoint structure independently of the authors, with disagreements resolved by majority vote (κ =0.71 viewpoints, κ =0.68 CDA-7); and evaluation never touched the gold file (Appendix C). The headline metrics are adapted from independent prior work rather than invented here: CR follows the claim-verification paradigm of SciFact (Wadden et al., 2020), VC follows the coverage framing of ExpertQA (Malaviya et al., 2024), and FS follows FActScore (Min et al., 2023).

What this does not settle. None of the above makes the benchmark neutral in the way an independently constructed benchmark would be, and the one clean test—a disagreement-aware system built by another group and evaluated on EVIRAG-BENCH by that group—is not something we can run ourselves. We release the gold annotations, adjudication records, and scoring scripts so that it can be run by others. We also take our own CCS result (Appendix F) as direct evidence that same-team metric design can miss a fundamental error for a long time, which is why we audit the full suite rather than trusting construct intuition, and why the Limitations section states a prediction in advance rather than reporting only results already in hand.

F Metric Audit

This appendix collects the metrics we discarded, the evidence for discarding them, and the validity checks we ran on the metrics we retained. Because one metric we designed turned out to measure the opposite of what we intended, we treat the whole suite as requiring validation rather than patching that metric alone.

F.1 Direction-validity audit

A metric is *direction-valid* if, on instances where the correct ordering is known independently of the metric, the metric orders them the same way. We tested every embedding-based metric in the paper against gold instances: query outputs for which the gold annotation fixes which response covers more of the viewpoint structure. We used 8 such instances drawn from 6 domains and 2 independent annotation sources, comparing a known-collapsed response against a known-multi-view response for the same query.

Metric	Uses gold?	Correct	Verdict
CCS	no	0/8	fails
SVC	yes	8/8	passes
ContradictionRecall	yes	8/8	passes
VC _{emb}	yes	8/8	passes

Table 15: Direction-validity audit on 8 gold instances (6 domains, 2 annotation sources). *Correct* counts instances where the metric orders the collapsed and multi-view responses in the direction the gold annotation requires. CCS is wrong on every instance (sign test $p=0.008$); the three reference-based metrics are correct on every instance.

The pattern in Table 15 is not noise, and its cause is structural rather than statistical. CCS is the only metric in the suite that never consults the gold annotation: it is computed entirely from the geometry of a response’s own claim clusters. A response that genuinely covers several separated viewpoints necessarily pushes its dominant cluster centroid away from the mean of its cluster centroids, which *raises* CCS — and CCS was reported as lower-is-better. The metric was therefore measuring internal cluster tightness and calling it viewpoint balance. Every metric that consults the reference passes. We draw the general lesson explicitly: reference-free geometric proxies for viewpoint structure should not be trusted without a direction check against gold instances.

F.2 SVC: a reference-based concentration metric

SVC (Eq. 1) measures the same construct CCS was intended to measure — concentration of a response on a single view — but against the gold view set rather than against the response’s own geometry, and by assignment rather than by cosine magnitude. Per-system values (lower is better): Vanilla RAG 0.783, Single-agent 0.731, MMR-RAG 0.647, PaperQA2 0.584, MADAM-RAG 0.531, EVIRAG-NoGraph 0.487, EVIRAG-NoMultiView 0.487, EVIRAG Full **0.287**. These values reproduce the direction the gold annotations require, the reverse of the ordering CCS induces; see Table 2 for the full per-system comparison.

F.3 VC_{kw}: a saturating sanity check

The keyword-overlap variant of ViewpointCoverage spans only 0.892–0.991 across all eight systems in Table 2, and MMR-RAG (0.991) scores above EVIRAG Full (0.984). A metric on which a 1998 retrieval heuristic and a seven-stage pipeline are separated by 0.007 is not measuring what distinguishes them: gold viewpoints are lexically recoverable from almost any response that mentions the topic, because coverage of a *keyword* is much weaker than coverage of a *position*. We therefore report VC_{kw} only as a check that a system has not omitted a viewpoint’s vocabulary entirely, never as evidence of viewpoint coverage. Per-system values: Vanilla 0.967, Single-agent 0.971, NoGraph 0.983, NoMultiView 0.892, MADAM-RAG 0.958, PaperQA2 0.942, MMR-RAG 0.991, EVIRAG Full 0.984.

F.4 Measured CCS values

For completeness and auditability we report the CCS figures as computed: Vanilla 0.571, Single-agent 0.565, NoGraph 0.553, NoMultiView 0.612, MADAM-RAG 0.549, PaperQA2 0.583, MMR-RAG 0.547, EVIRAG Full 0.518. Taken at face value these appear to favor EVIRAG; interpreted correctly they are uninformative, since the ordering they induce tracks cluster tightness rather than viewpoint balance. No claim in this paper rests on them.

F.5 Encoder-disjoint ViewpointCoverage

Table 16 recomputes VC_{emb} with `bge-base-en-v1.5` (Xiao et al., 2024), an encoder that appears nowhere in any system’s retrieval path.

System	MiniLM	BGE (disjoint)
Vanilla RAG	0.423	0.462
EVIRAG Full	0.847	0.781
Gap	+0.424	+0.319

Table 16: VC_{emb} under the shared retrieval encoder (`all-MiniLM-L6-v2`) and under a disjoint encoder (`bge-base-en-v1.5`). The gap shrinks by 0.105, which is our estimate of the shared-encoder advantage; the ordering is unchanged.

F.6 Confidence-tier stability under CDA-7 label perturbation

Because EVIRAG’s emitted confidence tier is derived in part from CDA-7 cause labels, and CDA-7 attribution is only 0.68 macro-F1 (Table 19), we tested how much the confidence output depends on getting those labels right. We perturbed the predicted CDA-7 label of every contradiction edge to its most-confusable neighbor — Methodological $\leftrightarrow$ Population and Statistical $\leftrightarrow$ Methodological, the two boundaries that dominate the confusion matrix — and recomputed the emitted tier. The tier is unchanged on 88.4% of queries; of the queries that move, almost all shift by one tier (high $\rightarrow$ medium or medium $\rightarrow$ low) and none cross from high to low. Confidence output is therefore substantially more robust than the underlying attribution accuracy, because the tier depends on the *presence and count* of contradiction edges and their source support more than on which of two adjacent causes is assigned. This does not make CDA-7’s 0.68 acceptable as an attribution result

Model	CR $\uparrow$	$VC_{emb}\uparrow$	CCS ‡
<i>Vanilla RAG</i>			
Qwen2.5-7B	0.541	0.391	0.589
Llama-3.1-70B	0.563	0.418	0.574
Qwen3.6-35B (main)	0.580	0.423	0.558
<i>EVIRAG Full</i>			
Qwen2.5-7B	0.648	0.703	0.547
Llama-3.1-70B	0.712	0.821	0.524
Qwen3.6-35B (main)	0.760	0.847	0.512

Table 17: Cross-backbone robustness on the education subset (50 queries). EVIRAG consistently outperforms vanilla RAG on CR and VC_{emb} across all three backbones. ‡ : as above, CCS is reported for auditability only and carries no direction (Appendix F).

— individual causal labels should be read as indicative — but it does mean the confidence signal does not inherit that error rate.

G Additional Robustness and Failure Analysis

Table 17 reports CR, VC_{emb} , and CCS for Vanilla RAG and EVIRAG Full across three backbone families on the 50-query education subset; EVIRAG outperforms vanilla RAG at every scale tested.

In the 22 queries where EVIRAG Full achieved $CR < 0.5$, the dominant pattern was NLI noise (9 cases), followed by claim-granularity mismatch (7), retrieval coverage gaps (4), and CDA-7 boundary confusion (2). These errors suggest that future gains are more likely to come from stronger verifiers and retrieval expansion than from replacing the overall disagreement-aware architecture.

CDA-7 classification accuracy. To evaluate whether the LLM-based CDA-7 module assigns correct disagreement causes, we compared system-predicted CDA-7 labels against the 250-query gold annotations. Table 19 reports per-class precision, recall, and F1. Macro-F1 is 0.68, indicating moderate but imperfect agreement; the main confusion occurs between Methodological and Population (boundary cases where sampling design effectively selects a different population) and between Statistical and Methodological (different analysis protocols producing different effect-size interpretations). These confusions mirror the boundary cases documented in our CDA-7 annotation guide and suggest that finer-grained NLI or a dedicated classifier could improve attribution accuracy.

Failure mode	Example	Why it fails	Likely fix
False contradiction	Related claims with different emphasis are labeled as conflict	Prompt-based NLI noise	Stronger cross-encoder verifier
Missing minority view	Rare terminology prevents retrieval of a dissenting paper	Retrieval coverage gap	Query expansion / terminology substitution
Over-amplified weak dissent	Single-source minority view receives undue salience	Output salience exceeds evidential weight	Stronger confidence display and source-count emphasis

Table 18: Representative failure modes observed during manual inspection of low-CR queries.

CDA-7 class	P	R	F1
Methodological	0.74	0.71	0.72
Population	0.65	0.62	0.63
Temporal	0.78	0.80	0.79
Operational	0.66	0.64	0.65
Statistical	0.62	0.58	0.60
Theoretical	0.71	0.68	0.69
Replication	0.70	0.66	0.68
Macro avg	0.69	0.67	0.68

Table 19: CDA-7 classification accuracy: system-predicted labels vs. gold annotations on 250 queries. Main confusions occur at the Methodological–Population and Statistical–Methodological boundaries.

H Qualitative Output Examples

Tables 20–21 show representative system outputs on three queries to concretely illustrate epistemic collapse. Both responses are fluent and locally accurate; the distinction is structural. The CCS values shown in these tables are retained for auditability only; as established in Appendix F they run in the wrong direction, and the homework example in Table 20 is where that is most visible. Read VC_{emb} and the view counts instead. Table 21 shows that the pattern is not specific to education. On the statin and minimum-wage queries vanilla RAG returns a single consensus statement ($VC_{emb}=0.33$ and 0.50) while EVIRAG recovers every gold viewpoint ($VC_{emb}=1.00$ in both cases), and the two queries receive different controversy classes—Stable for statins, Polarized for the minimum wage—so the synthesized output tracks the shape of the disagreement rather than applying one fixed multi-view template.

I Error Analysis: Full Breakdown

Manual inspection of the 22 queries where EVIRAG Full achieved $CR < 0.5$ reveals four failure categories; Table 18 (Appendix G) summarises representative cases.

NLI noise and false contradictions (41% of low-CR cases). Prompt-based NLI is the dominant

source of error. NLI models trained on general-domain benchmarks inherit annotation artifacts—surface cues such as negation words and hedging phrases that spuriously predict CONTRADICTS without genuine inferential conflict (Gururangan et al., 2018). Scientific claims routinely contain domain-specific hedging (“*may reduce*,” “*is associated with*,” “*in some populations*”) that triggers these artifacts. In addition, claims referring to the same outcome via different surface forms (e.g., “*reduces LDL*” vs. “*lowers cholesterol*”) may be mislabelled as contradictory rather than entailing, inflating graph density with spurious edges. Stronger cross-encoder verifiers fine-tuned on scientific text are the most direct mitigation (Wadden et al., 2020).

Claim granularity mismatch (32% of cases). Failures occur at the boundary between atomic and molecular claims (Gunjal and Durrett, 2024). Over-atomization—splitting a conditional claim such as “*Homework improves achievement for grades 7–12 but not primary school*” into two fragments—loses the qualifying structure that determines whether the pieces genuinely contradict a retrieved passage. Conversely, insufficiently decomposed claims bundle multiple assertions into one unit, making NLI verification unreliable. A related sub-failure is hedging stripping: LLM extractors omit qualifying phrases (“*in elderly patients*,” “*under controlled conditions*”) during decomposition, producing over-broad claims that NLI spuriously matches across distinct populations. This error type is causally upstream of NLI noise: malformed claim pairs enter the NLI module and compound the annotation-artifact problem.

Retrieval coverage gaps (18% of cases). Coverage failures arise from vocabulary mismatch—minority-view papers may use terminology that dense retrievers fail to bridge (e.g., “*net employment effect*” vs. “*job displacement*”)—and from retrieval-generation misalignment, where the gen-

Vanilla RAG	EVIRAG Full
<i>Query: Does homework improve academic achievement?</i>	
“Research consistently shows that homework improves academic achievement, with stronger effects for older students (grades 7–12). Meta-analyses report moderate positive correlations between homework time and academic performance. Homework builds self-regulation and reinforces classroom learning.” CCS: 0.38. VC_{emb}: 0.33 (1/3 views covered).	View 1 [Dominant, 6 sources, confidence: medium]: Conditional positive effect. Homework improves achievement for grades 7–12 ($d \approx 0.30$); effect near zero for primary grades. <i>CDA-7: methodological (grade-level scope differences).</i> View 2 [Alternative, 3 sources, confidence: medium]: Null meta-analytic effect when controlling for SES and teacher quality. <i>CDA-7: statistical + methodological.</i> View 3 [Minority, 2 sources, confidence: low]: High homework loads harm wellbeing with no academic benefit in high-achieving populations. <i>CDA-7: operational (achievement vs. wellbeing outcomes).</i> <i>Controversy class: Stable. CCS: 0.52. VC_{emb}: 1.00.</i>

Table 20: Homework query: Vanilla RAG collapses to the dominant conditional-benefit view; EVIRAG Full recovers all three ground-truth viewpoints with per-view source counts, CDA-7 causal attribution, and evidence-structured confidence.

Query	Vanilla RAG	EVIRAG Full (summary)
<i>Do statins prevent cardiovascular events in primary prevention?</i>	“Statins are recommended for high-risk primary prevention and reduce cardiovascular events.” (1 view; VC_{emb} : 0.33; CCS: 0.41)	View 1 [Dominant]: Benefit confirmed for high-risk groups (<i>CDA-7: population</i>). View 2: Marginal benefit with adverse-effect trade-off (<i>CDA-7: statistical</i>). View 3: Industry-bias concern undermining reported effect sizes (<i>CDA-7: methodological</i>). VC_{emb} : 1.00; Controversy class: Stable; CCS: 0.54.
<i>Does raising the minimum wage reduce employment?</i>	“Most studies find minimal employment effects from moderate minimum wage increases.” (1 view; VC_{emb} : 0.50; CCS: 0.44)	View 1 [Dominant]: Minimal aggregate employment effect (<i>CDA-7: methodological</i>). View 2: Significant job losses for low-wage workers in specific sectors (<i>CDA-7: population</i>). View 3: Positive employment via demand-side stimulus (<i>CDA-7: theoretical</i>). VC_{emb} : 1.00; Controversy class: Polarized; CCS: 0.50.

Table 21: Additional examples in biomedicine and economics. In both cases Vanilla RAG returns a single consensus statement that erases 2/3 of ground-truth viewpoints; EVIRAG Full provides structured multi-view output with per-view CDA-7 attribution and controversy-class labels.

eration stage ignores minority-view passages even when they are retrieved (Ru et al., 2024). A structural contributor is publication bias: null and negative results, which constitute the clearest evidence of scientific disagreement, are structurally underrepresented in academic corpora; the Skeptic agent cannot recover evidence that was never published or ingested.

CDA-7 attribution boundary confusion (9% of cases). CDA-7 macro-F1 = 0.68 (Table 19) reflects systematic confusion at three class boundaries: **Methodological vs. Population** (sampling design that selects a different cohort is classifiable as either); **Statistical vs. Methodological** (different analysis protocols producing different effect-size interpretations); and **Temporal vs. Replication** (a null result from a later study

is ambiguously either temporal supersession or replication failure). Binary boundary classifiers trained on each pair, rather than a single seven-way prompt, would reduce these confusions.